LAB ASSIGNMENT 8

From Code to Circuit: Building a 4-bit Mini-ALU and Program Counter

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1. GenAI Model Name

ChatGPT (GPT-6)

2. Prompt

Design a 4-bit Mini-ALU for addition and subtraction and a 4-bit Program Counter using Logisim Evolution. Explain the circuit components, connections, working principle, and verification with simple steps.

3. Response

A. 4-bit Mini-ALU

A 4-bit Mini-ALU performs addition and subtraction using two 4-bit inputs, A and B, and a mode input M.

Components Used:

- Four XOR gates
- One 4-bit adder
- Input pins A, B and M
- One 4-bit output pin S

Working Principle:

When M = 0, the circuit performs addition. When M = 1, it performs subtraction using the two's-complement method.

The Boolean operation is:

X = B XOR M

S = A + X + M

B. 4-bit Program Counter

A Program Counter stores the address of the next instruction. In this circuit, a 4-bit register stores the current count.

Components Used:

One 4-bit register

- One 4-bit adder
- One 4-bit multiplexer (MUX)
- Constant input 0001
- Clock, Enable and Reset inputs
- One 4-bit output pin PC

Working Principle:

The adder adds 0001 to the current count. The MUX selects either the current count or the incremented count. The Enable input controls the MUX selection, while the clock controls when the register updates. Reset clears the stored count according to the reset polarity of the register.

When Enable = 1, the counter increments on each active clock edge. When Enable = 0, the counter holds its current value.

4. Verification by Logisim

A. Mini-ALU Test Results

A	B	M	Expected S
0011	0010	0	0101
0101	0011	0	1000
0110	0010	1	0100
0100	0001	1	0011

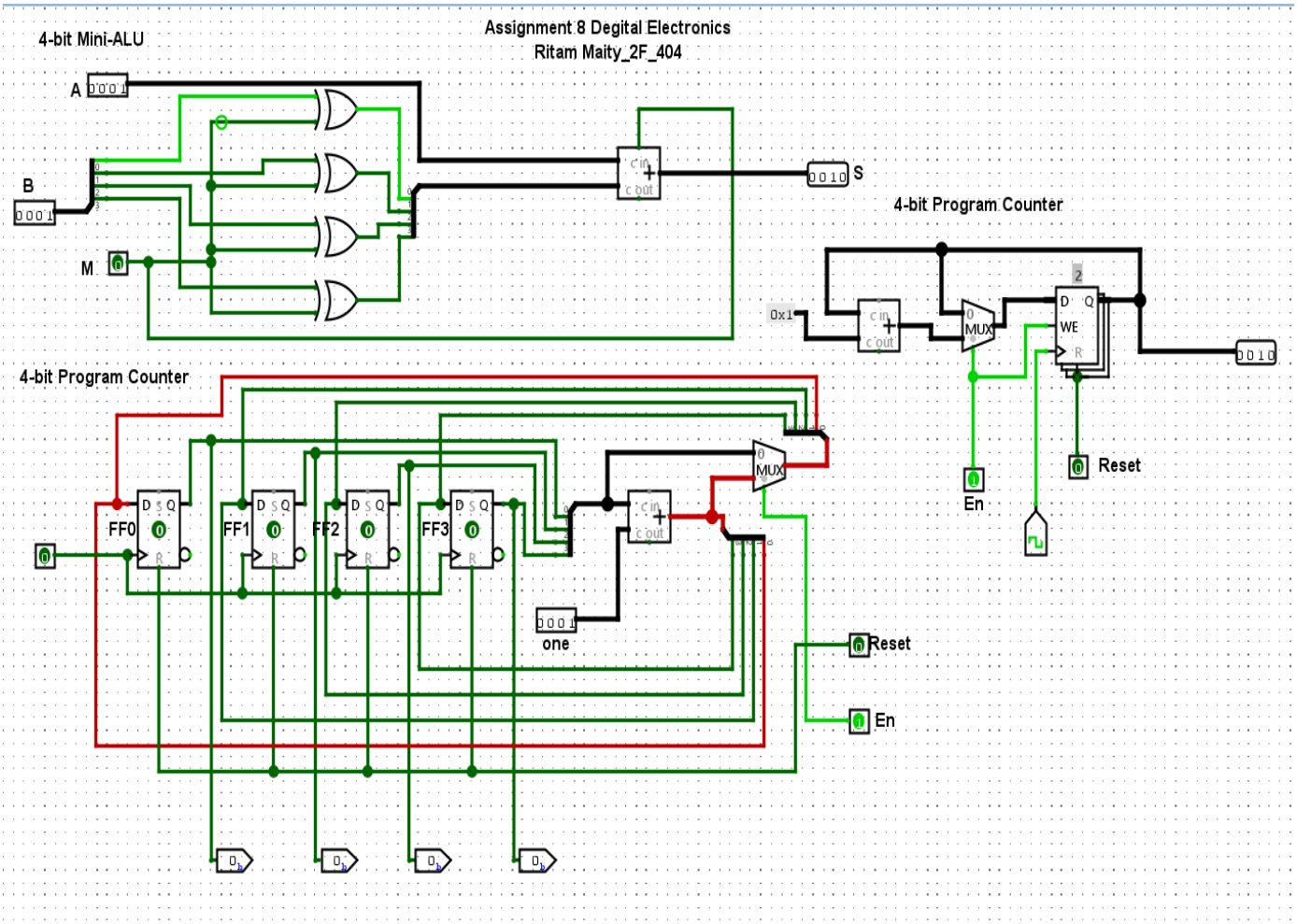
The circuit is considered correct when the observed outputs match the expected results.

B. Program Counter Test Results

Condition	Expected PC
Reset	0000
First clock pulse with Enable = 1	0001
Second clock pulse with Enable = 1	0010
Third clock pulse with Enable = 1	0011
Enable = 0	Holds the current value

The circuit is considered correct when the output follows the expected sequence.

5. Circuit Diagram :



6. Output

The Mini-ALU is designed to perform 4-bit addition and subtraction using a mode-select input. The Program Counter is designed to increment its stored value by one on each active clock edge when enabled.

7. Discussion

This assignment demonstrates how arithmetic operations and sequential circuits can be implemented using digital logic components. The Mini-ALU performs addition and subtraction, while the Program Counter stores and updates a 4-bit value. Logisim Evolution helps verify the circuit behavior through simulation.

8. GitHub Link

GitHub Repository: https://github.com/Ritam-Maity/Digital-Electronics-Lab